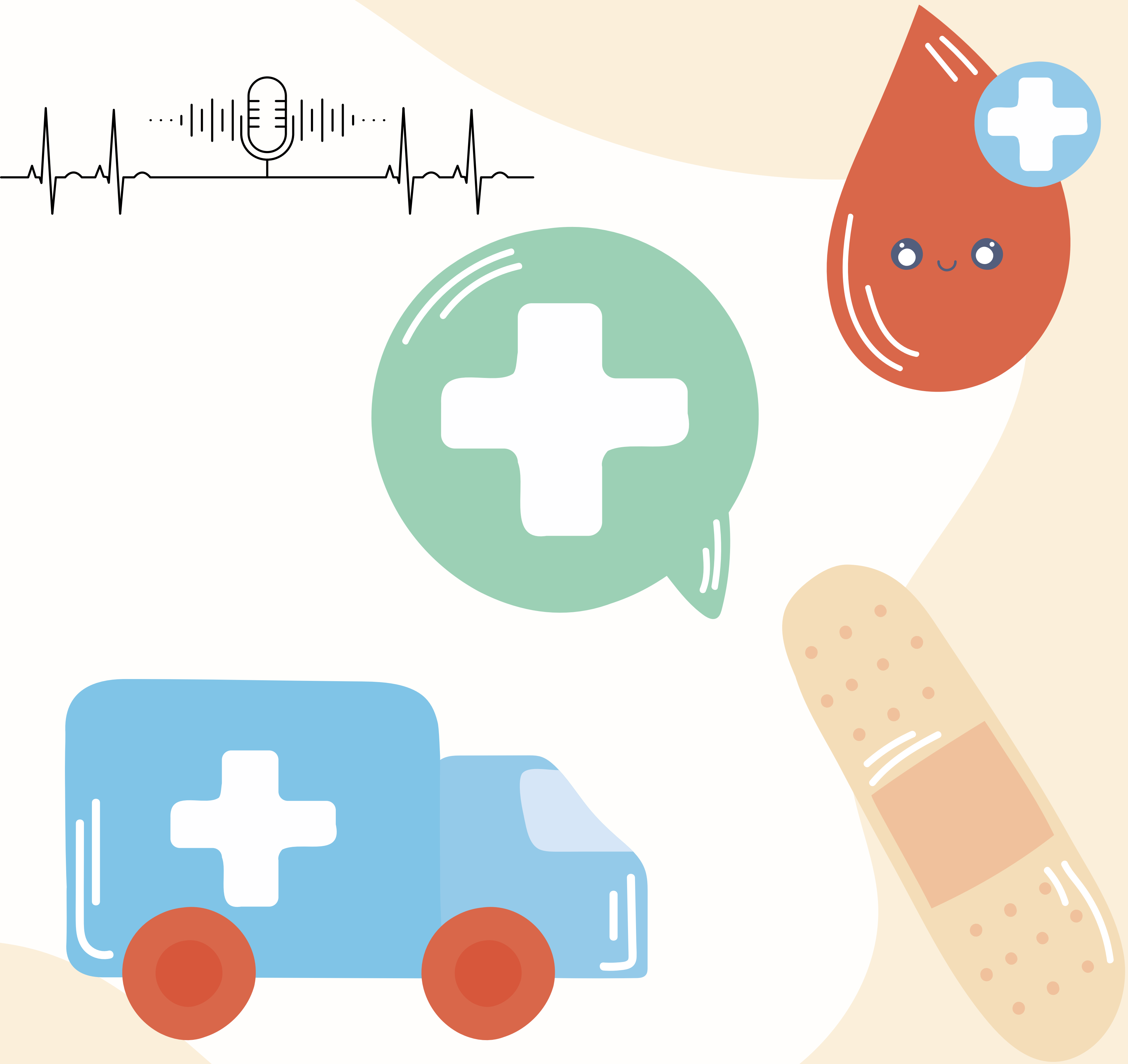
MedAlert



PREDICTING ED DECISIONS FROM EMS PRE-ARRIVAL REPORTS

Clean Text and Noisy ASR Robustness Evaluation

STUDENTS:

Nofar Refaeli ,Meital Gerasi

PROJECT REVIEW

MOTIVATING USE CASE

EMS pre-arrival handoff reports provide the Emergency Department with early clinical information before the patient arrives.

This information may help the ED prepare the appropriate care area and specialty consultation.

However, these reports are often brief, incomplete, variable in wording, and may contain ASR transcription errors.

PROBLEM DEFINITION

Task:
Predict ED decision labels from EMS pre-arrival report text.

input:
Synthetic EMS-to-ED handoff report

Outputs:
1. Arrival destination
2. Specialty consultation

Novelty:
Evaluate model performance on clean text and on noisy ASR transcripts created using TTS and ambulance-noise injection.

PREVIOUS WORK

Title	Assessing the Effectiveness of Automatic Speech Recognition Technology in Emergency Medicine Settings: A Comparative Study of Four AI-powered Engines , 2024	Deep learning-based in-ambulance speech recognition and generation of prehospital emergency diagnostic summaries using LLMs, 2025	Speech recognition technology in prehospital documentation: A scoping review, 2025
Task	Assess ASR performance for automatic clinical documentation in fast-paced and noisy EMS settings.	Generate emergency diagnostic summaries from in-ambulance speech using noise-robust speech recognition and LLMs.	Review how speech recognition technology is used or perceived in prehospital documentation, and identify paramedic user requirements.
Methods	Compared four ASR engines: Google Speech-to-Text Clinical Conversation, OpenAI Speech-to-Text, Amazon Transcribe Medical, and Azure Speech-to-Text	First, noisy ambulance audio was transcribed into text using a noise-resistant ASR model. Then, LLMs generated emergency diagnosis summaries from the transcript.	Scoping review using PRISMA-ScR and JBI methodology. Databases were searched for studies published between 2014 and March 2024.
Data	40 high-fidelity EMS simulation recordings, annotated into 23 EMS EHR categories	200 hours of audio: 18h real ambulance audio, 4h ambulance noise, and 178h AISHELL-1 Mandarin speech. Split: 70% train, 20% validation, 10% test.	8 included studies met the criteria; all were published from 2020 onward and were mostly small lab/simulation or focus-group studies.
Results	Google ASR performed best. Evaluation used Precision, Recall, F1 and semantic-similarity Accuracy. Critical fields were still weak: treatment F1=0.650 and medication F1=0.577.	Metrics: CER, API, RPI, FSS, and completion time. DeNoiseformer reached 52.92% CER on real ambulance audio. Qwen2.5-7B-Instruct with Stylized Prompt performed best. EMR completion time decreased from 20 to 14 min.	Speech recognition in paramedicine is still limited. Key requirements: hands-free use, noise reduction, battery life, and word accuracy. More live testing and user-centered design are needed.
Relation to our project	The study shows that ASR systems may fail to capture clinically relevant EMS information, especially treatment and medication details, in dynamic and noisy prehospital settings.	Both projects use noisy ambulance speech and ASR. Their output is a diagnostic summary; our output is ED decision labels.	speech recognition may improve paramedic documentation and handover, but noisy prehospital environments and transcription accuracy remain major challenges. Our project directly evaluates noisy ASR transcripts from ambulance handoff reports.

DATASET PREPROCESSING

- **Source:** MIMIC-IV-Ext CDS task-specific files merged by stay_id.
- **Files used:** triage_level.csv, diagnosis.csv, and specialty_referral.csv.
- **Working subset:** 2,139 complete cases with HPI, initial vitals, triage, diagnosis, and specialty referral.

Cleaned noisy diagnosis/specialty fields:
specialties 591 → 63,
diagnoses 1,886 → 1,797

Parsed initial vitals from text into numeric variables and removed clinically implausible values.

Labeling

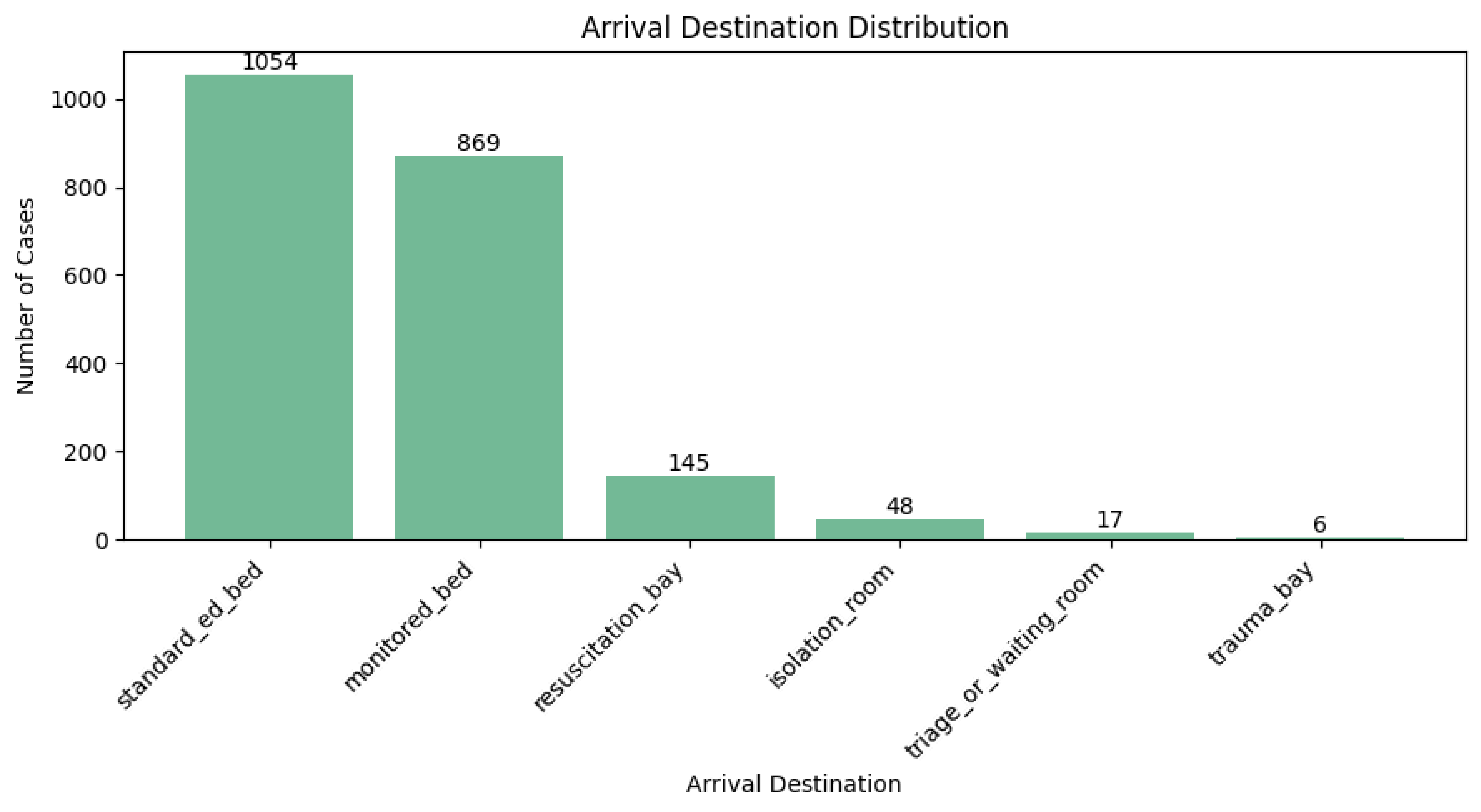
FIRST LABEL: ARRIVAL_DESTINATION

- Created using rule-based weak labeling.
- Input signals for labeling: clinical narrative, triage level, and parsed vital signs.
- Rules identify clinical instability, isolation risk, trauma/high-risk injury patterns, and low-acuity presentations.
- Used as target only; excluded from the text-generation prompt.

6 Final classes:

standard_ed_bed, monitored_bed, resuscitation_bay, isolation_room, trauma_bay, triage_or_waiting_room

TARGET LABEL DISTRIBUTION AND CLASS IMBALANCE



Labeling

SECOND LABEL: SPECIALTY_CONSULT_LABEL

• Derived from the cleaned specialty field.

• Original specialties were grouped into broader clinical consult labels.

• 63 original specialties were reduced to **18 final consult classes.**

• Generic labels such as medicine_or_ed_consult and other_consult were used only when no specific consult label was available.

• Used as target only; excluded from the text-generation prompt.

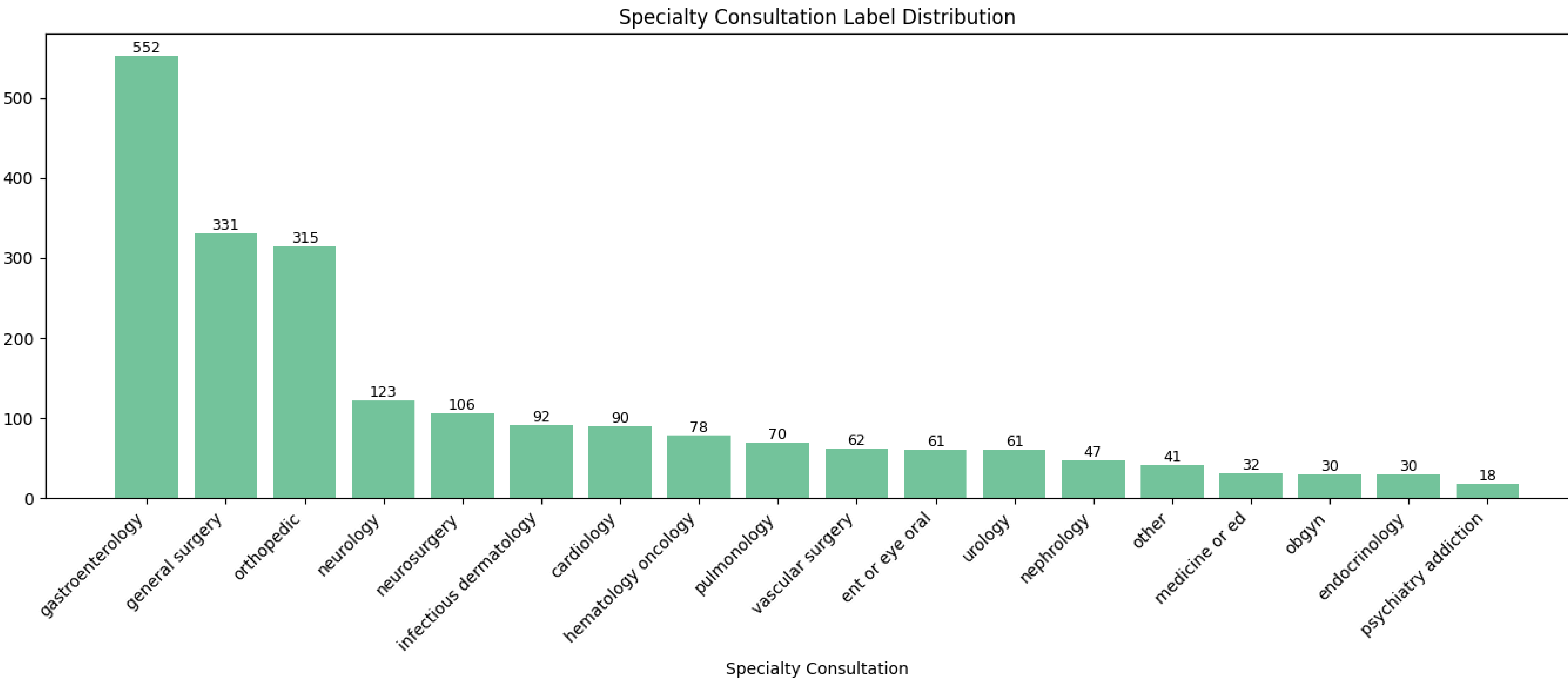
Examples:

orthopedics / hand surgery / podiatry → orthopedic_consult

ENT / ophthalmology / oral surgery / dentistry → ent_or_eye_oral_consult

**We intentionally prioritize ASR robustness over multi-label modeling in order to keep the experimental question focused and interpretable.*

TARGET LABEL DISTRIBUTION AND CLASS IMBALANCE



LEAKAGE-AWARE CASE SEED FOR REPORT GENERATION

- Each case was converted into a structured seed containing HPI, patient information, and initial vital signs.
- To reduce leakage, the seed excluded target labels, specialty fields, diagnosis fields, and triage, since these were related to the target-label creation.


```
{
  "source_case_id": 32822973,
  "patient_info": "Gender: Female, Race: WHITE, Age: 80",
  "hpi": "Patient is an ___ year-old patient with history of Sjogren's\nsyndrome, moderate MR, recent hospi
  "initial_vitals_raw": "Temperature: 100.2, Heartrate: 95.0, resprate: 28.0, o2sat: 100.0, sbp: 99.0, dbp:
  "vitals": {
    "temperature": 100.2,
    "heart_rate": 95.0,
    "respiration_rate": 28.0,
    "oxygen_saturation": 100.0,
    "systolic_bp": 99.0,
    "diastolic_bp": 47.0
  }
}
```

SYNTHETIC EMS REPORT GENERATION

2,139 LABELED CLINICAL CASES

↓

LEAKAGE-AWARE CASE SEEDS

↓

GPT-4.1-MINI TEXT GENERATION

↓

4 EMS REPORT VARIANTS PER CASE

↓

8,556 SYNTHETIC EMS REPORTS

4 VARIANTS PER CASE:

1. professional_complete
2. brief_radio_missing_details
3. patient_reported_uncertain
4. distracted_or_disorganized_handoff

FINAL DATASET LABEL DISTRIBUTION

**The generated dataset preserves the original class imbalance, since each source case produced four report variants.*

```
Specialty consult label distribution:
specialty_consult_label
gastroenterology_consult            2208
general_surgery_consult             1324
orthopedic_consult                  1260
neurology_consult                    492
neurosurgery_consult                 424
infectious_dermatology_consult       368
cardiology_consult                   360
hematology_oncology_consult          312
pulmonology_consult                  280
vascular_surgery_consult             248
ent_or_eye_oral_consult              244
urology_consult                      244
nephrology_consult                   188
other_consult                        164
medicine_or_ed_consult               128
obgyn_consult                        120
endocrinology_consult                120
psychiatry_addiction_consult          72
```

```
Arrival destination distribution:
arrival_destination
standard_ed_bed          4216
monitored_bed            3476
resuscitation_bay         580
isolation_room            192
triage_or_waiting_room     68
trauma_bay                 24
```

Example: Same Clinical Case, Different EMS Report Variants

```
Source case ID: 30001856
Arrival destination: standard_ed_bed
Specialty consult label: gastroenterology_consult

--------------------------------------------------------------------------------
Text variant: professional_complete
--------------------------------------------------------------------------------
We are en route with a 54-year-old male with a history of alcoholic cirrhosis and hepatitis C,
status post TIPS procedure, presenting with three weeks of worsening epigastric pain now
constant and radiating around his abdomen. He reports vomiting this morning, possibly with
blood, and subjective fever and chills earlier today. No chest pain or shortness of breath.
Vital signs are stable with a temperature of 99.1, heart rate 83, respiratory rate 16, oxygen
saturation 97%, and blood pressure 140/71. Patient remains awake and alert.

--------------------------------------------------------------------------------
Text variant: brief_radio_missing_details
--------------------------------------------------------------------------------
54-year-old male with history of cirrhosis and hepatic encephalopathy, presenting with
persistent epigastric pain radiating around the abdomen. Vomited this AM, possible blood noted.
Vitals stable, BP 140/70, O2 sat 97%, no respiratory distress. Transporting alert, pain
ongoing.
```

POST-PROCESSING AND LEAKAGE REDUCTION

After generation, reports were scanned for hospital-based or leakage-prone wording.

DETECTED PATTERNS

- Imaging/lab results
- Specialist consult wording
- Hospital workflow phrases

METHOD

Regex-based candidate detection → minimal LLM rewrite → second-pass check.

RESULTS

- Round 1 candidates: 461 reports (5.39%)
- Round 2 remaining candidates: 168 reports (1.96%)

POST-PROCESSING AND LEAKAGE REDUCTION

The goal was to preserve clinical meaning while keeping the text EMS-style.

```
--- ORIGINAL TEXT ---
51-year-old male with history of esophageal cancer post-esophagectomy, presenting with 2
days of worsening crampy abdominal pain and bilious vomiting. CT shows small bowel
obstruction. Vitals stable with BP 102 systolic, O2 sat 95% on room air. Transporting now.

--- CLEANED TEXT ---
51-year-old male with history of esophageal cancer post-esophagectomy, presenting with 2
days of worsening crampy abdominal pain and bilious vomiting consistent with small bowel
obstruction. Vitals stable with BP 102 systolic, O2 sat 95% on room air. Transporting now.
```

TRAIN / VALIDATION / TEST SPLIT

Split strategy:

• Group-based split by source_case_id: all 4 variants of the same clinical case were kept in the same split
• Class coverage check: all target classes are represented in train, validation, and test.

**This prevents leakage between train, validation, and test*

Final split:

70% TRAIN:

5,988 reports (1,497 source cases)

15% VALIDATION:

1,284 reports (321 source cases)

15% TEST:

1,284 reports (321 source cases)

ASR ROBUSTNESS EVALUATION SETUP

GOAL

- After selecting the best model for each target, evaluate its robustness on the same fixed test set:
- Clean text reports
- Noisy ASR transcripts after speech synthesis, ambulance noise, and Whisper transcription

https://drive.google.com/file/d/12f6D7XoTyqm1V-QM2hh8oOpsJEE8EI6Z/view?usp=drive_link

Test reports

↓

TTS conversion
(edge-tts, male voice, speed ×1.3)

↓

Clean audio

↓

Ambulance noise injection
(siren + white noise, medium level)

↓

Noisy audio

↓

Whisper transcription

↓

Noisy ASR text

↓

Clean text test set vs Noisy ASR test set

ASR Error Example: Clean Report vs Noisy Transcript

Mean WER: 0.474≈ 47%

WER measures the word-level difference between the clean EMS report and the ASR transcript.

In this example, about 31.6% of the words were changed, deleted, or inserted after noisy ASR transcription.

```
Original cleaned text:
We are transporting an 80-year-old female with a history of Sjogren's syndrome, moderate mitral
regurgitation, and recent sepsis from C. diff colitis complicated by respiratory failure
requiring intubation. She presents with worsening shortness of breath and cough for one day.
Vitals are temperature 100.2, heart rate 95, respiratory rate 28, blood pressure 99/47, and
oxygen saturation 100% on 2 liters nasal cannula. She denies chest pain, nausea, vomiting, or
abdominal pain. Patient is stable but tachypneic.

ASR noisy transcript:
We are transferring an 80 year old female with a history of children's syndrome moderate
microrecordination and recent census from C. Diffleitis complicated by respiratory failure
requiring innovation. She presents the first thing Jordan's of Breffin Club for Monday. High
older temperature 100.2, heart rate 95, respiratory rate 28, blood pressure 99.7, an oxygen
saturation 100% on two liters nasal cannula. She denied Chess D, nausea, vomiting or a humble
pain. Haysha is stable but to keep you.
```

MODELS

TF-IDF + LOGISTIC REGRESSION BASELINE

BASELINE APPROACH:

- **Text representation:** TF-IDF
- **Classifier:** Logistic Regression
- Hyperparameter tuning was performed on the validation set
- Final evaluation was performed on the held-out test set

TUNED HYPERPARAMETERS:

arrival_destination:

ngram_range = (1, 2)

min_df = 2

class_weight = balanced

specialty_consult_label:

ngram_range = (1, 1)

min_df = 1

class_weight = balanced

TF-IDF + LOGISTIC REGRESSION BASELINE: RESULTS

arrival_destination

Classification report:

	precision	recall	f1-score	support
isolation_room	0.56	0.25	0.35	36
monitored_bed	0.63	0.67	0.65	504
resuscitation_bay	0.33	0.15	0.21	84
standard_ed_bed	0.74	0.80	0.77	640
trauma_bay	0.00	0.00	0.00	12
triage_or_waiting_room	0.00	0.00	0.00	8

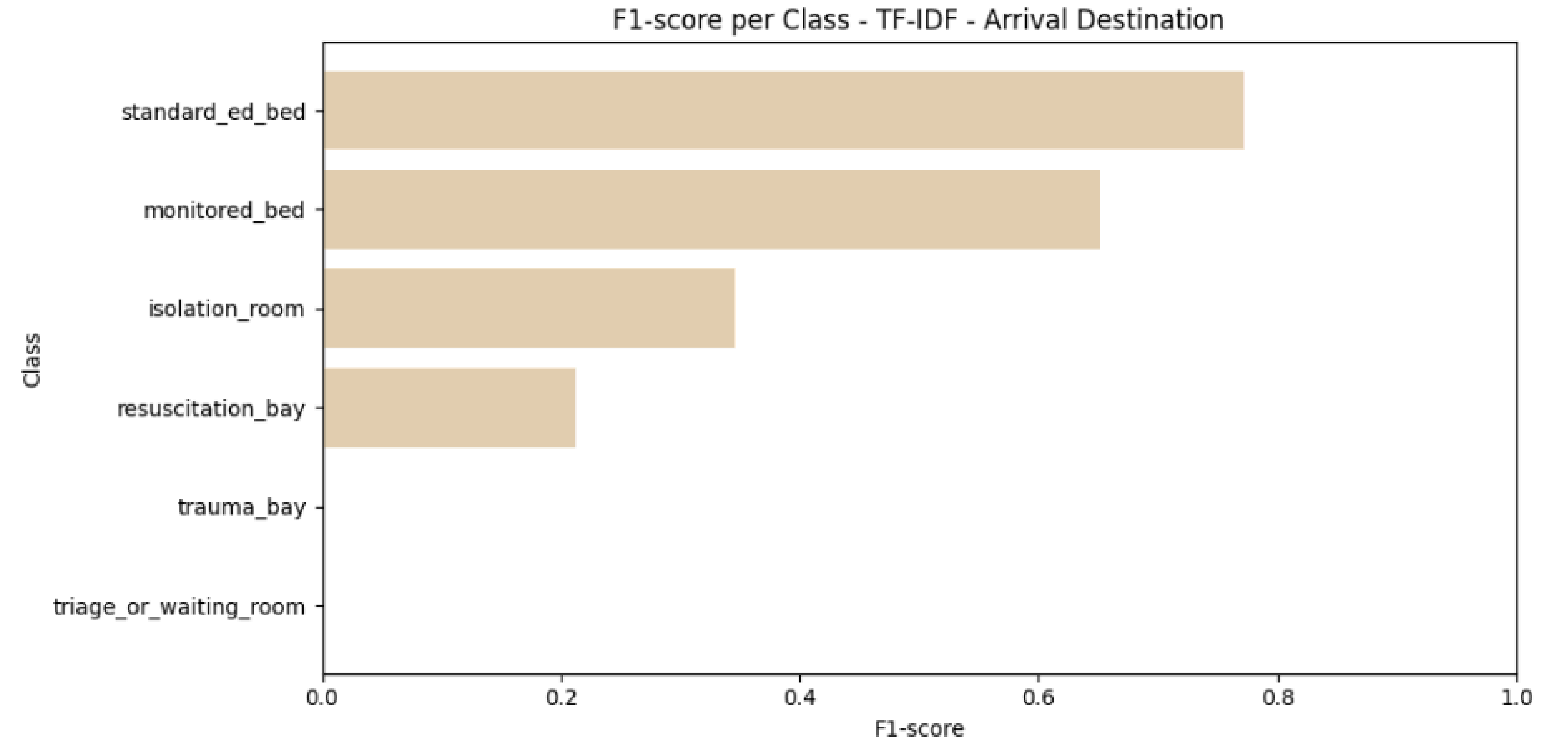
Accuracy = 68.1%

Macro F1 = 0.330

Weighted F1 = 0.664

TF-IDF + LOGISTIC REGRESSION BASELINE: RESULTS

arrival_destination



TF-IDF + LOGISTIC REGRESSION BASELINE: RESULTS

specialty_consult

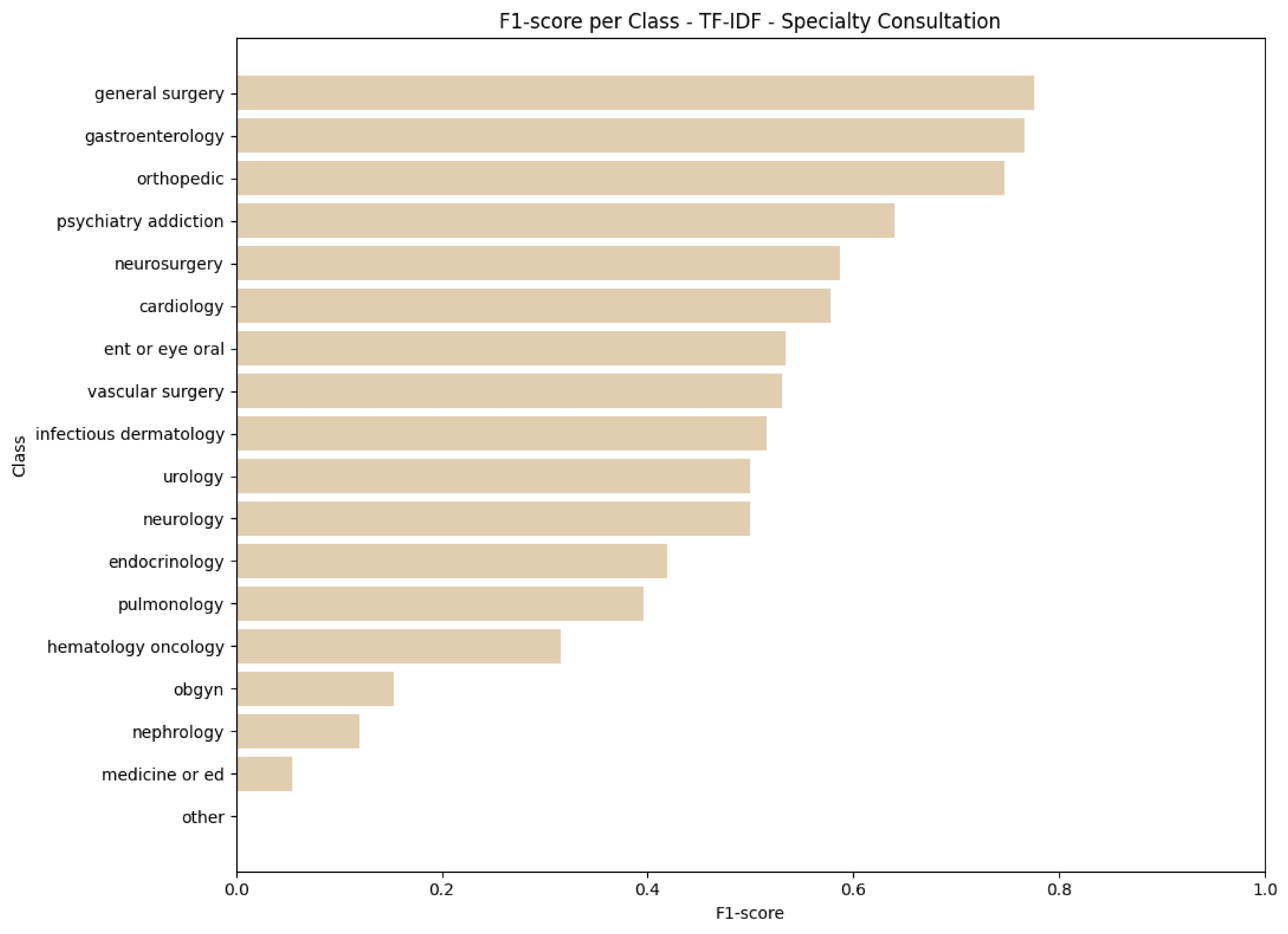
Accuracy = 61.9%
Macro F1 = 0.452
Weighted F1 = 0.622

Classification report:

	precision	recall	f1-score	support
cardiology_consult	0.54	0.62	0.58	60
endocrinology_consult	0.47	0.38	0.42	24
ent_or_eye_oral_consult	0.51	0.56	0.53	48
gastroenterology_consult	0.79	0.74	0.77	336
general_surgery_consult	0.78	0.77	0.78	188
hematology_oncology_consult	0.31	0.32	0.32	28
infectious_dermatology_consult	0.47	0.57	0.52	56
medicine_or_ed_consult	0.08	0.04	0.05	24
nephrology_consult	0.12	0.12	0.12	24
neurology_consult	0.49	0.51	0.50	92
neurosurgery_consult	0.69	0.51	0.59	92
obgyn_consult	0.16	0.15	0.15	20
orthopedic_consult	0.76	0.74	0.75	160
other_consult	0.00	0.00	0.00	8
psychiatry_addiction_consult	0.47	1.00	0.64	8
pulmonology_consult	0.35	0.45	0.40	40
urology_consult	0.50	0.50	0.50	40
vascular_surgery_consult	0.47	0.61	0.53	36

TF-IDF + LOGISTIC REGRESSION BASELINE: RESULTS

specialty_consult



Transformer Model Setup

BioClinicalBERT

Input:

text_cleaned EMS report

Task:

Multi-class text classification

- Tokenization with max_length = 128
- 3 epochs
- batch_size = 16
- learning_rate = 2e-5
- Best model selected by validation Macro F1
- Regular Trainer, without class weights

BioClinicalBERT Results: arrival_destination

Classification report:

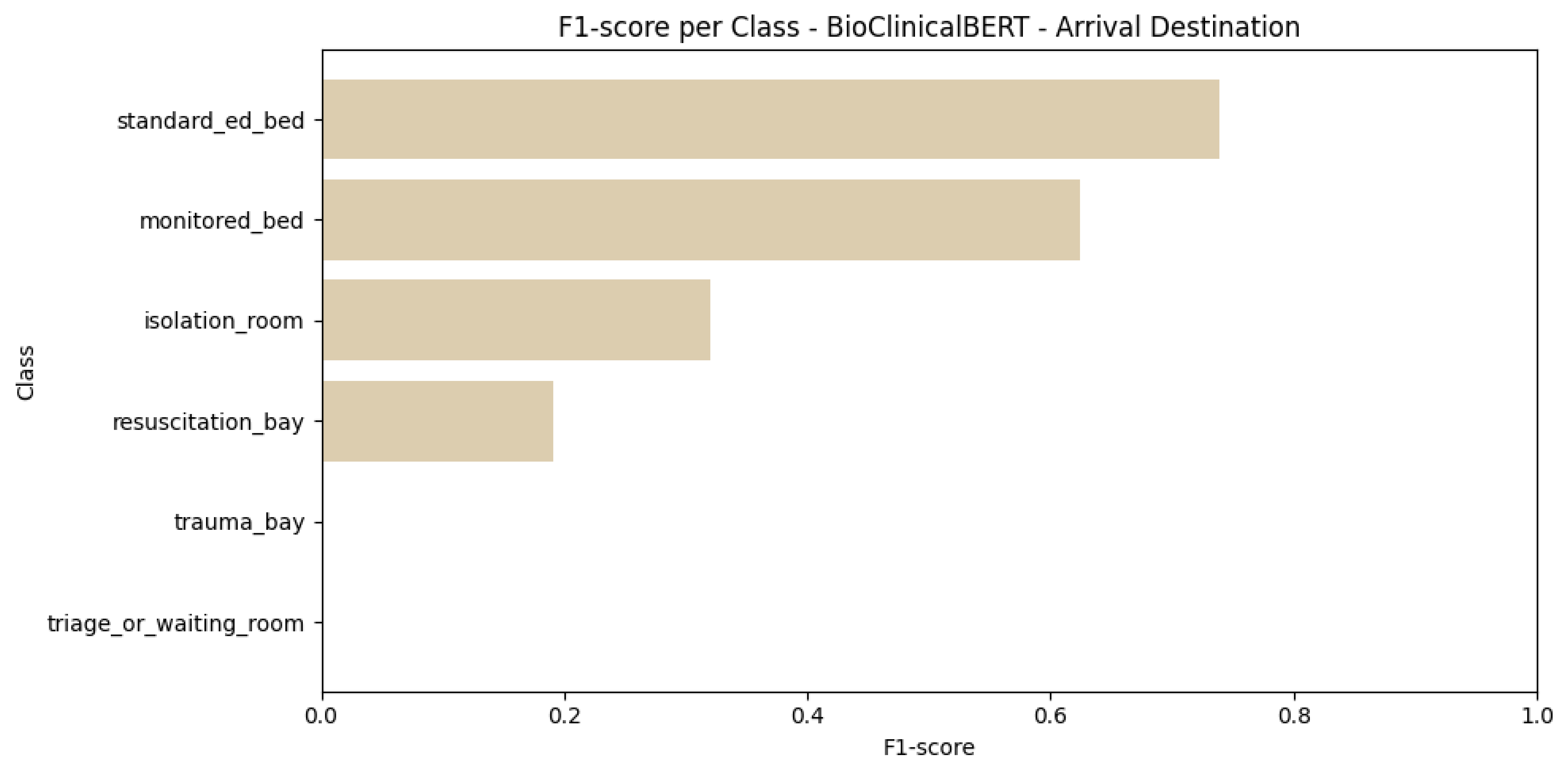
	precision	recall	f1-score	support
isolation_room	0.57	0.22	0.32	36
monitored_bed	0.58	0.67	0.62	504
resuscitation_bay	0.25	0.15	0.19	84
standard_ed_bed	0.74	0.73	0.74	640
trauma_bay	0.00	0.00	0.00	12
triage_or_waiting_room	0.00	0.00	0.00	8

Accuracy = 64.7%

Macro F1 = 0.312

Weighted F1 = 0.635

BioClinicalBERT Results: arrival_destination



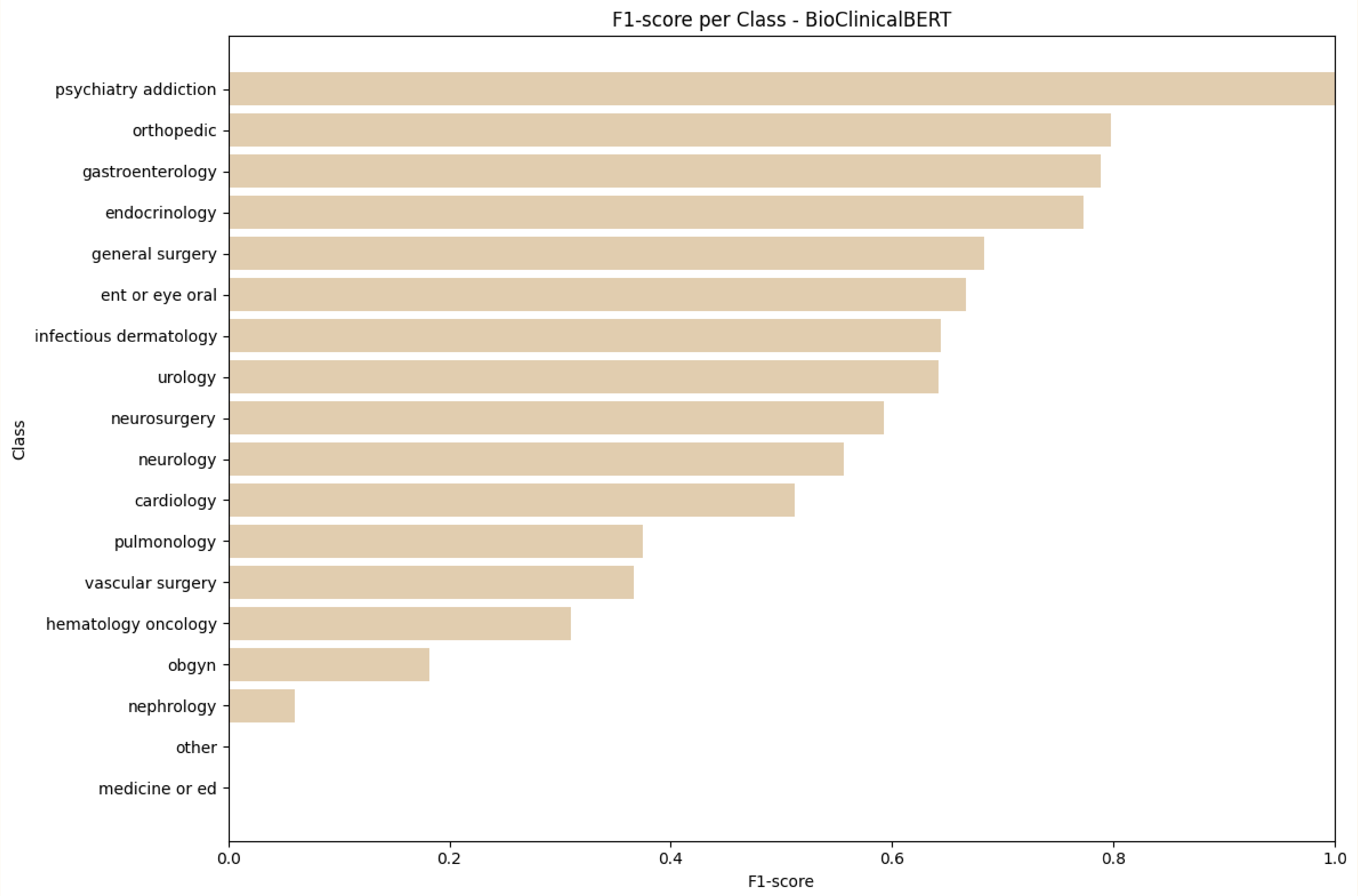
BioClinicalBERT Results: specialty_consult

Accuracy = 65.7%
Macro F1 = 0.497
Weighted F1 = 0.638

Classification report:

	precision	recall	f1-score	support
cardiology_consult	0.48	0.55	0.51	60
endocrinology_consult	0.85	0.71	0.77	24
ent_or_eye_oral_consult	0.69	0.65	0.67	48
gastroenterology_consult	0.73	0.85	0.79	336
general_surgery_consult	0.69	0.68	0.68	188
hematology_oncology_consult	0.30	0.32	0.31	28
infectious_dermatology_consult	0.61	0.68	0.64	56
medicine_or_ed_consult	0.00	0.00	0.00	24
nephrology_consult	0.11	0.04	0.06	24
neurology_consult	0.55	0.57	0.56	92
neurosurgery_consult	0.69	0.52	0.59	92
obgyn_consult	1.00	0.10	0.18	20
orthopedic_consult	0.75	0.85	0.80	160
other_consult	0.00	0.00	0.00	8
psychiatry_addiction_consult	1.00	1.00	1.00	8
pulmonology_consult	0.38	0.38	0.38	40
urology_consult	0.63	0.65	0.64	40
vascular_surgery_consult	0.37	0.36	0.37	36

BioClinicalBERT Results: specialty_consult



Target Label	Model	Accuracy	Macro F1	Weighted F1
arrival_destination	**TF-IDF + Logistic Regression**	**0.681**	**0.330**	**0.664**
	BioClinicalBERT	0.647	0.312	0.635
specialty_consult	TF-IDF + Logistic Regression	0.619	0.452	0.622
	BioClinicalBERT	**0.657**	**0.497**	**0.638**

PLAN TOWARD FINAL SUBMISSION

Clean vs ASR: Evaluate the best trained models on clean text and noisy ASR transcripts

Model tuning: Try more epochs, lower learning rate

Improve rare class performance: using class weights.
Controlled oversampling / Text augmentation.

Perform error analysis using confusion matrices.